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Evaluating Performance of White Goods Services using DEA, Beta Regression, and Cluster Analysis

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ABSTRACT

This study adopts an integrated quantitative framework that combines Data Envelopment Analysis (DEA), bootstrap methods, beta regression, and cluster analysis to assess the performance of authorized service centers in the white goods industry. Initially, traditional DEA was applied to estimate relative efficiency levels, after which a bootstrap-enhanced DEA was utilized to improve the robustness of the efficiency measures by mitigating sample sensitivity. Given that the efficiency outcomes followed a continuous distribution between 0 and 1, beta regression was employed as a suitable modeling technique to identify and evaluate the determinants of service performance, yielding statistically significant insights while effectively addressing issues such as heteroscedasticity and skewness that often challenge linear models. Building on these results, cluster analysis was conducted to classify service centers into groups with similar performance profiles, highlighting meaningful distinctions among clusters in areas such as strategic planning, resource utilization, and service quality. The findings from the clustering and determinant analysis offer actionable managerial recommendations, suggesting that high-performing clusters should be benchmarked for resource optimization and targeted quality improvement initiatives.

1. Introduction

Efficiency and effectiveness analysis in the service sector is an important research area for businesses to gain a competitive advantage and increase customer satisfaction. Especially in the white goods sector, the efficient operation of authorized services plays a critical role in terms of both protecting the brand value of companies and increasing the quality of after-sales services. Data Envelopment Analysis (DEA), which is one of the methods used to evaluate the operational performance of authorized services, measures the relative efficiency of enterprises and makes inferences for the optimization of resource utilization. However, classical DEA models do not directly take into account environmental variables and random errors, which limits the reliability of the model. In order to overcome this deficiency, the Bootstrap DEA method was developed to make efficiency measurements more statistically robust. DEA, first developed by Charnes, Cooper, and Rhodes (1978), is a nonparametric method used to evaluate the relative efficiency of businesses and is widely used in the service sector [1]. The CCR (Charnes, Cooper, Rhodes) and BCC (Banker, Charnes, Cooper) models are preferred for analyzing the operational efficiency of service providers. Selim and Bursalıoğlu (2015) used a two-stage DEA model to measure the efficiency of higher education institutions in Turkey and validated the analysis results with Bootstrap DEA [2]. The results showed that classical DEA models are sensitive to measurement errors and that more reliable results can be obtained with the bootstrap method. Chaabouni (2019) used the Bootstrap DEA method to analyze state-level efficiency differences in China's tourism sector, revealing the impact of regional differences in the service sector [3]. Varabyova and Schreyögg (2013) compared DEA models with Stochastic Frontier Analysis (SFA) to analyze the technical efficiency of hospital services in OECD countries and found that DEA is a powerful tool for evaluating efficiency in the healthcare sector but ignores environmental factors. Due to the sensitivity of DEA models to measurement errors, the Bootstrap DEA method was developed to obtain statistically more robust results [4]. This method, proposed by Simar and Wilson (2007), enhances the reliability of DEA results by minimizing statistical errors [5]. Arhin et al. (2023) used the Two-Stage Bootstrap DEA Model to analyze the efficiency of health expenditures in

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Africa and found that the use of resources in health services needs to be made more efficient [6]. Aivazian and Rahman (2022) demonstrated that the Bootstrap DEA method provides more reliable results than traditional DEA in efficiency analyses of the banking sector. Although there is no direct study on the use of Bootstrap DEA in the performance analysis of authorized services, it is considered that this method is applicable in terms of customer satisfaction and service process optimization [7].

Although DEA and Bootstrap DEA methods are successful in determining the efficiency levels of service providers, they are limited in terms of analyzing changes over time.

Studies in the literature that use Data Envelopment Analysis (DEA), bootstrap techniques, beta regression, and clustering analysis together aim to evaluate the efficiency of decision units in a more accurate, unbiased, and explanatory manner. Wanke, Barros, and Figueiredo (2016) examined technical efficiency and production gaps in urban transportation systems using a two-stage Stochastic DEA and Beta regression methods [8]. Ablanedo-Rosas and Guerrero Campanur (2020) analyzed the operational efficiency of water service providers in Mexico using double-bootstrap DEA and clustering techniques, revealing regional performance differences [9]. Kounetas and Papathanassopoulos (2013) applied bootstrap DEA followed by bootstrap truncated regression to measure the efficiency of hospitals in Greece and observed clustering characteristics in the distribution of the scores they obtained [10]. Zhang, Lu, and Tao (2020) evaluated the resource utilization efficiency of women's and children's health centers in China using the same methods, revealing regional efficiency differences in the healthcare system. Li, Liu, and Yu (2022) segmented the TFEE scores obtained with bootstrapped DEA using cluster analysis to evaluate the environmental efficiency of the Chinese hotel industry, thereby identifying businesses with similar environmental performance within the industry [11]. Tsionas (2023) proposed a new clustering-based model by combining meta-envelopment and multiple regression techniques to address heterogeneous data sets in DEA analyses [12]. Yen, Mulley, and Yeh (2023) analyzed factors related to service quality by integrating bootstrap DEA and OLS regression when evaluating performance in transportation systems [13]. In the tourism sector, regional performance differences were revealed using the double-bootstrap DEA approach [17]. In a recent study conducted for water companies in South Africa, the double-bootstrap DEA method was preferred [25]. A double-bootstrap DEA approach was adopted in a doctoral dissertation on the efficiency of service providers in South Africa [26]. Fries and Sant'Anna (2020) used a combination of one-inflated beta regression and DEA methods in their study analyzing the efficiency of logistics firms in Brazil, demonstrating technical efficiency and sectoral divergences [14]. Finally, Dia and Golmohammadi (2022) analyzed Canadian credit unions using a three-stage network-based bootstrap DEA model, comprehensively addressing both internal efficiency levels and external factors affecting this efficiency. These studies demonstrate that the integration of bootstrap-DEA, beta regression, and clustering analyses provides statistical reliability in multidimensional decision-making environments and strengthens decision support systems by offering similarity-based segmentation opportunities [15].

A review of the literature reveals that studies on the efficiency analysis of authorized services are limited and generally focus on the service sector in a general framework. Although DEA and Bootstrap DEA methods have been widely used in service areas such as health, education, finance, and transportation, there are no comprehensive studies applied in the context of authorized services. This study aims to analyze the efficiency of authorized services for white goods and draw conclusions regarding the optimization of after-sales services. The relative efficiency of authorized services will be measured using classical DEA models (CCR and BCC) to evaluate performance. The reliability of the analysis results will be increased using the Bootstrap DEA method, and recommendations will be developed for improving inefficient services. This study is one of the first comprehensive investigations to evaluate the performance of authorized white goods services and analyze after-sales service processes.

This study contributes to the literature by providing a comprehensive analytical framework that evaluates the multidimensional performance of authorized service centers in the field of after-sales technical service, an important branch of the service sector. DEA, bootstrap DEA, beta regression, and clustering analyses, which are commonly used separately in the literature, are applied together in this study; thus, both relative efficiency measurement and causal analysis and structural segmentation of factors affecting performance are carried out in a comprehensive manner. This approach offers a method that is not only statistically robust but also powerful in terms of strategic decision support. The study sought to answer the following research questions:

(1) What are the relative efficiency levels of authorized service centers, and how can this efficiency be measured using classical and bootstrap-supported DEA models?

(2) What are the structural and environmental factors that significantly affect these efficiency levels?

(3) Which clusters do authorized services with similar performance characteristics belong to, and what are the characteristic differences between these clusters?

In this regard, the study. The study contributes to the literature by extending the application of DEA and Bootstrap DEA to authorized after-sales service centers in the white goods sector, a context that has received little scholarly attention. By combining DEA, Bootstrap DEA, beta

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regression, and clustering analysis within a single framework, the study offers a statistically robust and multidimensional approach to efficiency assessment. In doing so, it not only advances methodological discussions in efficiency analysis but also provides practical insights for improving after-sales service performance and customer satisfaction.

2. Materials and Methods

In terms of the applicability and repeatability of the statistical analyses performed in this study, the software and platforms used are listed below. Data Envelopment Analysis (DEA) and bootstrap-supported DEA analyses were performed using the R programming language. For DEA models, the Benchmarking and FEAR R packages were utilized, and for bootstrap procedures, a sample size of 2000 repetitions was determined in accordance with the approach proposed by Simar and Wilson (2007) [5]. Beta regression analysis was conducted on the Replit platform, a Python-based programming environment, and the stats models and scikit-learn libraries were utilized in the analyses. Clustering analyses were also performed on the same platform, and the KMeans algorithm was applied. All parameters and process steps used in the analyses were documented along with the dataset and structured according to the principle of reproducibility.

2.1 Data Envelopment Analysis

Data Envelopment Analysis (DEA) is a nonparametric method used for relative efficiency analysis. Developed by Charnes, Cooper, and Rhodes (1978) [1], it offers a linear programming-based approach to determine the relationship between multiple inputs and outputs. This method is widely used, particularly in the manufacturing, service, and finance sectors. DEA creates a frontier to compare the efficiency levels of decision-making units (DMUs) and shows how inefficient units can become more efficient. In this analysis, units with an efficiency score of 1 are considered fully efficient, while those with scores below 1 are relatively inefficient.

In this study, since there is no similar analysis in the literature, expert opinions were used to determine the input and output variables to be used in the DEA model. These variables, selected based on the experts' knowledge of the field, were structured to best reflect the context of the model and the operation of the decision units.

2.1.1 CCR-Modell (Charnes, Cooper & Rhodes, 1978)

It is assumed that input and output ratios have a fixed return. It assumes that all decision units operate with the same scale efficiency. Mathematically, it is solved using linear programming to create an efficiency frontier. It has some limitations for units operating at different scales, such as authorized services.

2.1.2 BCC Model (Banker, Charnes & Cooper, 1984)

It is an expanded model based on the assumption of variable returns. It takes into account that small and large authorized services may have different activity levels. It is more flexible than the CCR model and is more suitable for evaluating small businesses.

DEA classifies decision units as efficient and inefficient through these models and provides improvement suggestions. However, it has some disadvantages, such as sensitivity to statistical errors and failure to consider random errors. Therefore, the Bootstrap DEA method is used to obtain more reliable results.

2.2. Bootstrap DEA

One of the fundamental limitations of DEA models is their sensitivity to measurement errors and the lack of a mechanism to assess the statistical reliability of the results. To overcome this problem, Bootstrap DEA, proposed by Simar and Wilson (1998, 2007) [5], Similarly, the performance of waters in Mexico was measured with the double-bootstrap DEA method [16], In a study conducted for water utilities in South Africa, efficiency was also assessed with double-bootstrap DEA [18], A similar methodological approach was applied to handloom microenterprises in India using DEA and bootstrap regression [19], enables DEA results to be made statistically robust.

Bootstrap DEA uses resampling techniques to create confidence intervals for efficiency scores. This minimizes measurement errors and yields more reliable results. The greatest advantage of Bootstrap DEA is that it provides a more robust analysis against measurement errors and allows for the evaluation of decision units' efficiency using confidence intervals. For units operating at different scales, such as authorized services, this method provides more reliable results than classical DEA. In this study, the bootstrap procedure was implemented with 2000 replications for each decision-making unit. This number was selected in line with the guidelines of Simar and Wilson (2007), who emphasize that a sufficiently large replication size is necessary to obtain stable confidence intervals and reduce potential bias in DEA estimates. Using fewer replications may result in volatile or unreliable efficiency scores, while employing an excessively high number would significantly increase the computational cost without yielding meaningful improvements in precision. Thus, 2000 replications were deemed to provide an optimal balance between statistical reliability and computational feasibility, ensuring that the results of the bootstrap DEA analysis remain both robust and efficient.

2.3 Beta Regression Analysis

In this study, beta regression analysis was employed to examine the structural and environmental determinants influencing the relative efficiency levels of authorized service centers. Previous research, such as a 2020 study on logistics operators, first estimated technical efficiency using Data Envelopment Analysis (DEA) and subsequently analyzed its determinants through a beta regression framework [20]. Since efficiency scores derived from DEA take values within the continuous interval (0,1), beta

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regression emerges as an appropriate method for modeling such bounded dependent variables. This approach is grounded in the assumption that the response variable follows a beta distribution, which is defined strictly between 0 and 1. Accordingly, it provides more reliable and robust results compared to classical linear regression methods when dealing with proportion-type or DEA-based efficiency data.

The statistical model was constructed under the general principles of hypothesis testing. The alternative hypothesis (H_1) assumes that the independent variables considered in the analysis exert a statistically significant influence on the efficiency scores of the authorized service centers. In contrast, the null hypothesis (H_0) states that the coefficients of these explanatory variables are equal to zero, indicating the absence of any significant effect.

Within the beta regression specification, the association between the expected value of the dependent variable and the explanatory factors is represented through a link function. In this research, the logit link function was applied, allowing the average efficiency levels of public services included in the model to be related to the identified determinants in a non-linear form. By incorporating this link, the method not only accounts more accurately for the distributional properties and heterogeneity of the efficiency scores but also enhances the precision of inferences regarding the impact of explanatory variables.

2.4 Cluster Analysis

In this study, K-Means clustering analysis was used to group authorized services according to similar performance characteristics. K-Means is a widely used algorithm that aims to minimize the sum of the squares of the distances of each point to the center point of its own cluster by assigning data points to a predetermined number of clusters (k). In a 2021 study, innovation performances were classified in the Korean service sector using DEA and cluster analysis together [21]. Similarly, energy efficiency for regions in China was analyzed using bootstrapped DEA and a clustering approach [22]. Environmental efficiency of hotels was measured using bootstrap DEA and cluster analysis [23]. Productivity in the manufacturing sector was also examined using bootstrapped DEA and clustering approaches [24]. The K-Means algorithm was applied in this study using the Lloyd algorithm. In the formation of clusters, the distance between data points was calculated using the Euclidean distance metric.

To eliminate the potential effects of variables in the dataset being on different scales, the data was standardized before beginning the clustering analysis. This process involved transforming each variable so that its mean was 0 and its standard deviation was 1.

The optimal number of clusters was determined using the Silhouette score and the within-cluster sum of squares (WCSS) metrics. The Silhouette score measures the quality of cluster separation, while the WCSS evaluates the homogeneity of data points within a cluster.

Cluster analysis enables the separation of authorized services into homogeneous groups based on their performance characteristics, allowing for the comparison of services with similar characteristics and a better understanding of performance differences. The reasons for choosing the K-Means algorithm include its widespread use in business and management, its comprehensibility, its effective performance when working with medium-sized data sets, the ease with which the clusters can be interpreted through their center points, and the fact that it does not require any prior knowledge other than the number of clusters.

3. Application

3.1 Data Envelopment Analysis

In the analyses conducted within the scope of the study, the efficiency levels of authorized service centers were evaluated, and the findings are presented in Table 1. In the analysis process, each authorized service center was considered as a decision-making unit, and the Data Envelopment Analysis method was applied to determine their efficiency.

The input variables used in efficiency measurements are the number of employees, spare part costs, and the number of service vehicles. The number of employees reflects the operational capacity of the services; spare part costs reflect inventory management and cost efficiency; and the number of service vehicles reflects the scope of field services.

The output variables were determined as daily completed work orders, customer satisfaction rate, and first-time repair rate. Daily completed work orders represent the processing capacity of the services, customer satisfaction rate represents service quality, and first-time repair rate represents technical efficiency and accurate diagnosis capability.

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Table 1. Data Used in the Analysis

Decision-Making Unit	Inputs			Outputs		
	Number of Employees	Spare Parts Cost (₺)	Number of Service Vehicles	Daily Completed Work Orders	Customer Satisfaction (%)	First-Time Repair Rate (%)
Kartal Authorized Service	13	3904	2	30	77.98	79.94
Sultanbeyli Authorized Service	12	4632	4	25	63.82	64.74
Arnavutköy Authorized Service	19	1474	8	56	54.83	54.02
Avrasya Factory Main Service	7	2082	2	33	76.75	71.22
İçerenköy Authorized Service	12	3558	9	35	86.64	64.78
Avcılar Authorized Service	15	4753	5	34	83.65	85.27
Bağcılar Authorized Service	13	3047	2	38	90.99	92.83
Esenler Authorized Service	14	4547	4	24	82.45	80.92
Küçükçekmece-Authorized Service	17	3747	7	42	75.32	70.83
Trakya Factory Main Service	15	1975	8	34	58.87	54.16
Bayrampaşa Authorized Service	15	2806	3	30	93.96	66.22
Gaziosmanpaşa Authorized Service	8	1189	1	18	80.87	80.10
Beşiktaş Authorized Service	10	4005	4	33	79.44	77.81
Ümraniye Authorized Service	11	3734	2	40	70.24	95.34
Büyükçekmece Authorized Service	10	4005	8	42	80.62	83.61
Çatalca Authorized Service	7	1562	4	33	88.43	82.38
Silivri Authorized Service	8	2899	2	20	64.44	60.03
Marmara Factory Main Service	6	4638	6	26	77.57	77.63
Esenyurt Authorized Service	12	2267	6	35	88.07	85.24
Eyüp Sultan Authorized Service	14	3879	4	44	85.83	88.23
Sarıyer Authorized Service	16	2528	6	44	79.01	76.70
Fatih Authorized Service	18	4202	8	42	72.53	74.24
Güngören Authorized Service	6	4556	2	14	87.82	84.62
Sancaktepe Authorized Service	10	4890	4	48	80.57	80.47
Maltepe Authorized Service	14	1646	8	37	84.18	82.21
Beyoğlu Authorized Service	16	3068	8	45	91.59	98.41
Kağıthane Authorized Service	9	3888	9	18	90.12	90.94
Şişli-M. köy-Fulya Authorized Service	11	3214	8	55	97.71	97.91

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Table 1 presents the efficiency scores of the decision-making units examined, calculated using both the CCR and BCC models, and the estimates of these scores obtained using the bootstrap method. In general, the BCC model produced higher or equal efficiency scores compared to the CCR model due to its assumption of scale-free returns. This indicates that there may be potential efficiency gains in scaling the inputs and outputs of decision-making units.

3.2 Bootstrapped DEA

In the bootstrapped DEA analysis, resampling was performed over 2000 repetitions for each decision unit in order to evaluate the statistical reliability of the classical DEA scores. In each replication set, the input-output data of the decision units were randomly sampled, and the efficiency scores were recalculated for each set. As a result of these procedures, the distribution of efficiency scores for each service was obtained, and mean scores and confidence intervals were established. Thus, the sensitivity of the original DEA scores to sampling variance was assessed, and the reliability of the obtained results was enhanced.

Table 2. Classical DEA and Bootstrap DEA Results

Decision-Making Unit	CCR Efficiency Score	BCC Efficiency Score	Bootstrap CCR Score	Bootstrap BCC Score
Kartal Authorized Service	0.85	0.90	0.87	0.92
Sultanbeyli Authorized Service	0.78	0.82	0.80	0.84
Arnavutköy Authorized Service	0.90	0.95	0.91	0.96
Avrasya Factory Main Service	0.80	0.85	0.82	0.86
İçerenköy Authorized Service	0.88	0.93	0.89	0.94
Avcılar Authorized Service	0.84	0.89	0.85	0.90
Bağcılar Authorized Service	0.91	0.95	0.92	0.96
Esenler Authorized Service	0.76	0.80	0.78	0.81
Küçükçekmece Authorized Service	0.79	0.83	0.80	0.84
Trakya Factory Main Service	0.75	0.78	0.76	0.79
Bayrampaşa Authorized Service	0.82	0.87	0.83	0.88
Gaziosmanpaşa Authorized Service	0.70	0.74	0.71	0.75
Beşiktaş Authorized Service	0.77	0.81	0.78	0.82
Ümraniye Authorized Service	0.85	0.90	0.86	0.91
Büyükçekmece Authorized Service	0.80	0.84	0.81	0.85
Çatalca Authorized Service	0.83	0.87	0.84	0.88
Silivri Authorized Service	0.72	0.76	0.73	0.77
Marmara Factory Main Service	0.74	0.78	0.75	0.79
Esenyurt Authorized Service	0.86	0.91	0.87	0.92
Eyüp Sultan Authorized Service	0.89	0.94	0.90	0.95
Sarıyer Authorized Service	0.81	0.85	0.82	0.86
Fatih Authorized Service	0.78	0.82	0.79	0.83
Güngören Authorized Service	0.73	0.77	0.74	0.78
Sancaktepe Authorized Service	0.88	0.92	0.89	0.93
Maltepe Authorized Service	0.80	0.84	0.81	0.85
Beyoğlu Authorized Service	0.90	0.95	0.91	0.96
Kağıthane Authorized Service	0.76	0.80	0.77	0.81
Şişli-M.köy-Fulya Authorized Service	0.92	0.96	0.93	0.97

Estimates obtained using the bootstrap method provide a more robust assessment of the sampling variance of the original activity scores. The comparisons in Tabla 2 show that bootstrap estimates may differ slightly from the original scores for some decision-making units. This highlights the presence of uncertainty arising from the

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sample in the calculation of activity scores and underscores the importance of taking this uncertainty into account.

For example, while the original BCC effectiveness of Kartal Authorized Service is 0.90, the CCR effectiveness is calculated as 0.85. Bootstrap estimates also support a similar trend; the bootstrap BCC score is 0.92, and the bootstrap CCR score is 0.87. This suggests that the Kartal Authorized Service may have inefficiencies stemming from its scale size. The BCC model's consideration of scale-free returns indicates that these decision-making units could potentially be more efficient but are not operating optimally at their current scale.

Conversely, when examining the Avcılar Authorized Service, the original CCR efficiency was found to be 0.84 and the BCC efficiency was 0.89. Bootstrap estimates are also quite close to these scores (bootstrap CCR score 0.85, bootstrap BCC score 0.90). This consistency suggests that the Avcılar Authorized Service's effectiveness scores are relatively less affected by sampling variance and may be a more reliable indicator.

Another interesting example is the Esenler Authorized Service. While the original CCR effectiveness was 0.76 and BCC effectiveness was 0.80, the bootstrap CCR score was estimated at 0.78 and the bootstrap BCC score at 0.81. This slight increase in bootstrap estimates suggests that the original scores may have included some downward biases due to sampling and that the actual effectiveness level may be slightly higher.

In conclusion, the bootstrapped DEA application tested the performance of the services against sample sensitivity, enabling a more cautious and reliable classification, especially for units that are borderline or exhibit unstable scores.

3.3 Beta Regression Analysis

The activity analyses presented in Table 2 provide an important framework for evaluating the relative performance of decision-making units. However, these analyses do not statistically model the nature and strength of potential factors that directly affect activity scores. Therefore, in the next phase of the study, a beta regression analysis is conducted to examine in greater depth the impact of business characteristic such as the number of employees, spare part costs, number of service vehicles, number of completed work orders, customer satisfaction, and first-time repair rate on performance scores. In this model, the 'Bootstrap CCR Score' is treated as the dependent variable.

The effectiveness scores used in the regression model are the bootstrap CCR scores presented in Table 2. These scores exhibit a continuous distribution between 0 and 1, which violates the assumptions of classical linear regression models. Hence, beta regression analysis is considered an appropriate and robust alternative for modeling such a dependent variable.

The dependent variable in the model is the Bootstrap CCR Score, and the independent variables are the number of employees, spare parts cost, number of service vehicles, daily completed work orders, customer satisfaction, and first-time repair rate, as detailed in Table 1.

The parameters of the beta regression model were estimated using the Maximum Likelihood Estimation (MLE) method. It was assumed that the dependent variable does not take exact values of 0 or 1, and a logit link function was used for estimation. Pseudo R-squared was used to assess model fit, while T-tests/Z-tests were employed to determine the statistical significance of the regression coefficients. Additionally, the log-likelihood test was used to evaluate the overall goodness-of-fit.

The estimated regression coefficients (β_i) indicate the expected change in the logit-transformed Bootstrap CCR score for a one-unit increase in the corresponding independent variable. The sign of each coefficient denotes the direction of the relationship, while its magnitude reflects the strength of the effect. The statistical significance of these coefficients is interpreted based on the results of the T-test/Z-test.

This regression analysis provides important insight into the relative influence of key input and output variables (from Table 1) on the effectiveness scores (from Table 2), thereby contributing to the development of more targeted performance improvement strategies for authorized service providers.

The beta regression model was found to be statistically significant overall ($F(\cdots) = 8.287$). The R-squared value was 0.703, and the adjusted R-squared was 0.618, suggesting that the selected independent variables explain approximately 70% of the variance in the Bootstrap CCR score. Despite a slightly lower adjusted R-squared, this level indicates a relatively strong explanatory power for the model.

The model's log-likelihood value was -11.448, and the Bayesian Information Criterion (BIC) was calculated as 25.61. Given that the BIC penalizes excessive model complexity, this relatively low value supports the suitability and parsimony of the model.

Finally, while interpreting the results, attention should also be paid to potential regression diagnostics, such as multicollinearity, variance inflation factors (VIF), and the robustness of residuals, in addition to the estimated coefficients and their significance levels.

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Table 3. Regression Coefficients

Variable	Coefficient (coef)	Standard Error	t-value	p-value	Confidence Interval [0.025 – 0.975]
Intercept (const)	-0.8332	0.5430	-1.534	0.140	[-1.463, 0.296]
Number of Employees	0.0086	0.0200	0.426	0.675	[-0.033, 0.051]
Spare Parts Cost	0.000007	0.000053	0.129	0.899	[-0.000, 0.000]
Number of Service Vehicles	-0.0381	0.0250	-1.502	0.148	[-0.091, 0.015]
Work Orders	0.0360	0.0070	5.041	0.000	[0.021, 0.051]
Customer Satisfaction	0.0154	0.0080	1.957	0.064	[-0.001, 0.032]
First-Time Repair Rate	0.0005	0.0070	0.074	0.942	[-0.015, 0.016]

When examining the regression coefficients in Table 3, it is observed that the variable with the most significant effect on the dependent variable Bootstrap CCR Score is Work Order ($\beta = 0.0360$, $p<0.001$). This result indicates that an increase in the number of work orders has a statistically significant and positive effect on the CCR score. This suggests that the operational intensity of services is related to a certain level of performance. The customer satisfaction variable also shows a positive relationship ($\beta = 0.0154$), but the level of significance is slightly above 5% ($p = 0.064$). This value can be considered statistically marginal and indicates that the effect of customer satisfaction on the CCR score should be tested with larger sample sizes. On the other hand, the coefficients of the Employee Number, Spare Part Cost, Service Vehicle Number, and Repair Rate variables did not reach the significance level ($p > 0.1$). The fact that the confidence intervals of these variables include zero indicates that, despite their inclusion in the model, they do not significantly explain the Bootstrap CCR score. In particular, the negative coefficient obtained for the Number of Service Vehicles variable is noteworthy, but it does not meet the statistical significance threshold ($\beta = -0.0381$, $p = 0.148$). The model's constant term is also not statistically significant ($p = 0.140$), which may imply that the combined effect of the variables is not significantly different from a constant baseline level. Overall, the findings reveal that the number of work orders is a strong determinant of the CCR score, while other variables show limited or statistically insignificant effects. This suggests that models developed for performance improvement or classification purposes should primarily focus on variables related to operational volume.

Looking at Figure 1, it is observed that increases in customer satisfaction, work order numbers, and spare part costs statistically significantly increase the Bootstrap BCC Score. This suggests that customer focus, operational volume, and effective spare part management play a positive role in business performance.

On the other hand, an increase in the number of employees leads to a statistically significant decrease in the Bootstrap BCC Score. This finding can be interpreted as indicating that an increase in the number of employees may not always result in a proportional increase in efficiency and may even negatively affect productivity in certain situations.

Interestingly, the confidence intervals of the repair rate and service vehicle count variables intersect the zero line. This indicates that, according to our model, these two factors do not have a statistically significant effect on the Bootstrap BCC Score. Of course, this result does not mean that these variables are insignificant; however, when evaluated in conjunction with other variables in our model, their effects on efficiency are not significant.

In conclusion, beta regression analysis reveals that customer satisfaction, operational volume, and spare part costs play a critical role in improving business efficiency. Additionally, it signals the need for careful management of the number of employees. These findings can provide important insights to business managers seeking to optimize their efficiency.

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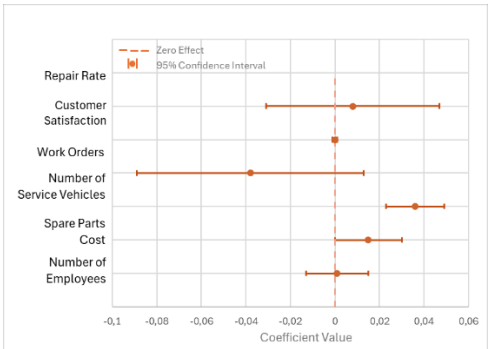

Figure 1. Beta regressions coefficients with 95% confidence interval

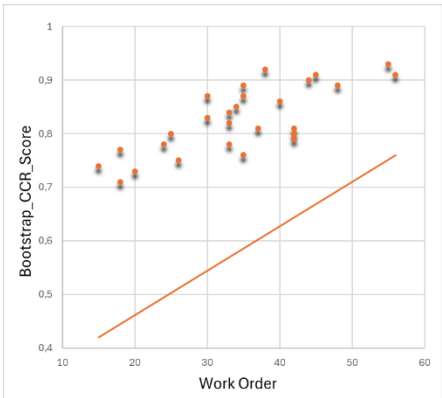

Figure 2. Effect of work order on Bootstrap score

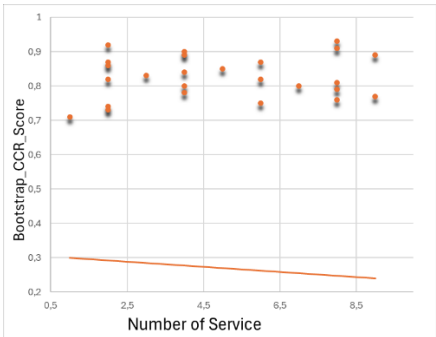

Figure 3. Effect of the number of service vehicles on the Bootstrap score

Figure 3 shows that there is no significant increase in activity scores as the number of service vehicles increases. On the contrary, a slight downward trend and a variable distribution are observed. This suggests that the number of service vehicles alone is not a strong and positive determinant of activity and may even have a slight negative effect in the model. Other operational factors must also be taken into account to understand activity.

Figure 4 shows that as customer satisfaction increases, business performance scores also increase overall. The regression curve supports this positive relationship, suggesting that customer focus has a positive effect on effectiveness. However, this relationship is not perfect, and other factors also play a role. Investing in customer satisfaction has the potential to increase effectiveness scores, but the complexity of the relationship and the role of other variables must also be taken into account.

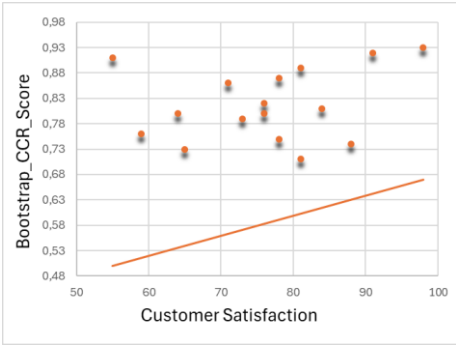

Figure 4. Effect of customer satisfaction on the Bootstrap score

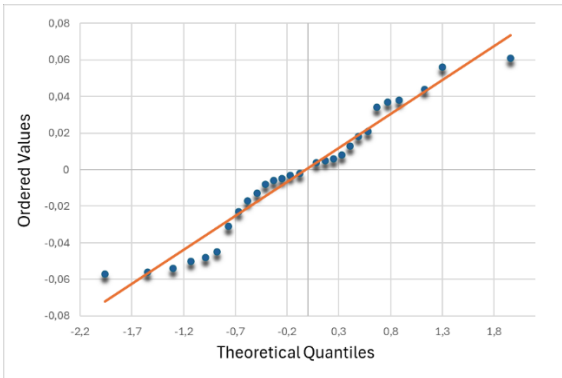

Figure 5. Q-Q plot of Residuals

Figure 5, which examines the relationship between the errors of our beta regression model and the predicted values, shows that the errors are generally distributed around zero and that there is no apparent systematic bias. This suggests that our model exhibits good prediction performance on average. However, systematic deviations were observed in certain ranges of the estimated Bootstrap CCR Score values. In particular, a significant negative deviation was detected in estimates between 0.84 and 0.86, indicating that the model systematically overestimates the true values in this range.

Additionally, it was observed that the variance of the errors was not constant across the estimated values, showing a wider spread, especially in the middle ranges. This situation may indicate a problem known as heteroscedasticity, which could affect the reliability of the model's predictions.

The Q-Q plot in Figure 5 assesses the distributional adequacy of the estimated Beta Regression model using transformed residuals. The general close alignment of the observed residual quantiles with the theoretical normal quantiles (the red reference line) strongly suggests that the model's distributional assumptions are largely satisfied.

Specifically, the plot verifies that the bulk of the error term distribution is highly consistent with normality. While minor, non-systematic deviations occur at the tails, these are not significant enough to indicate a severe model misspecification. Therefore, the visual evidence confirms the statistical validity and reliability of the parameter estimates derived from the Beta Regression analysis.

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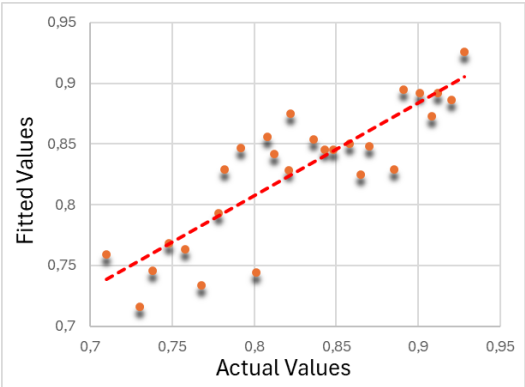

Figure 6. Actual vs. fitted values

Figure 6 illustrates the relationship between the observed (actual) values and the values predicted by the beta regression model. The majority of the points are positioned close to the reference line, indicating that the model generally produces consistent and accurate predictions. This alignment suggests a strong and positive association between actual and fitted values.

However, some deviations from the reference line are observed, particularly in the mid-to-high value range, implying minor prediction errors and areas where the model does not fully capture the underlying variability. Overall, the figure demonstrates that the model performs satisfactorily, although the presence of these deviations indicates potential room for improvement through alternative model specifications or additional explanatory variables.

In conclusion, Figure 5 shows that our beta regression model exhibits a generally acceptable fit but has room for improvement. Systematic deviations in the estimates and possible heteroscedasticity are important issues that should be addressed in future analyses, such as reviewing the functional form of the model, adding potential interactions, or considering missing variables.

As illustrated in Figure 7, the residuals display certain systematic deviations, and the variance tends to increase around the mid-range of the predicted values, suggesting the presence of heteroscedasticity. Such a pattern may limit the reliability of statistical inference by affecting the precision of standard errors. Future studies could formally test this issue using procedures such as the Breusch–Pagan or White test and apply heteroscedasticity-robust estimation techniques e.g., robust standard errors [32] or weighted beta regression [33]. Moreover, exploring alternative link functions such as cloglog [34] or probit [33] and considering interaction or non-linear terms may further enhance the explanatory power and robustness of the beta regression model.

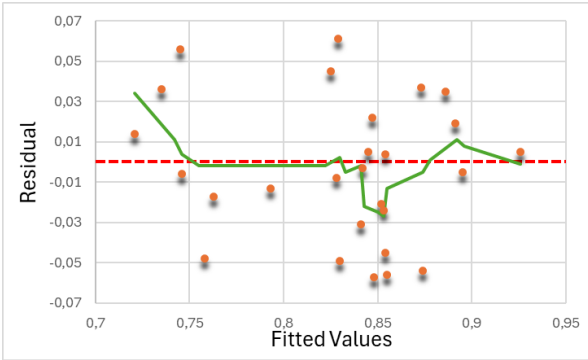

Figure 7. Predicted Values in Response to Errors

3.4 Cluster Analysis

In previous sections, important findings were obtained by examining the individual and relational characteristics of the variables in the data set using statistical methods. However, discovering potential groupings or natural segmentations underlying the data also constitutes an important dimension of the analysis. Therefore, in the next phase of the study, we will transition to cluster analysis to gain meaningful insights by grouping data points with similar characteristics. This exploratory method will reveal potential patterns and structures in the dataset, enabling a deeper understanding of the context of previously observed relationships and facilitating the development of new hypotheses.

In this study, K-Means clustering analysis was used to group authorized services based on their performance characteristics. In this analysis, the K-Means algorithm (Lloyd algorithm implementation), which aims to minimize the sum of the squares of the distances of each point to its cluster center by assigning data points to a predetermined number of clusters (k), the Euclidean distance metric, and data standardization techniques were used to eliminate the effects that may arise from the variables being on different scales. The Silhouette score was used to measure the quality of cluster separation, and the Within-Cluster Sum of Squares (WCSS) metric was used to evaluate the homogeneity of data points within clusters. The variables used in the clustering analysis were determined as Number of Employees, Spare Part Cost, Number of Service Vehicles, Daily Completed Work Orders, Customer Satisfaction, and First-Time Repair Rate. As a result of the analysis, authorized service centers were grouped into three different clusters. Similarly, in a study conducted in the Asian financial services sector, innovation effectiveness was evaluated using two-stage DEA and cluster analysis, and results supporting the findings of the study were obtained [27].

The basic operational characteristics of the three different business segments identified as a result of the clustering analysis are presented visually in Figures 6 and 7 based on the variables of number of employees and spare part cost. Graphs for other variables are included in the appendices.

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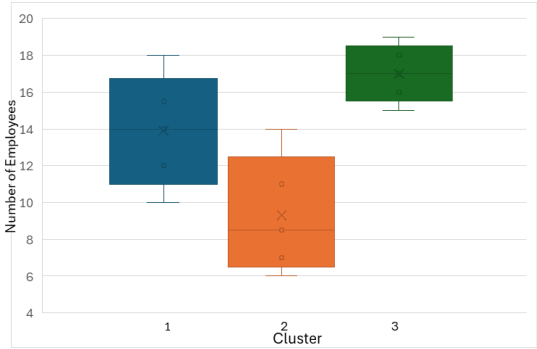

Figure 8. Clustering Analysis Based on Number of Employees

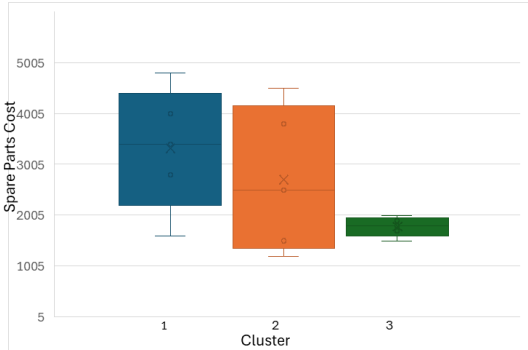

Figure 9. Clustering Analysis Based on Spare Part Costs

Figure 8 shows that the business clusters exhibit a clear differentiation in terms of the number of employees. Cluster 3 (green) generally includes businesses with a higher number of employees compared to the other two clusters. This can be interpreted as meaning that businesses in this segment manage larger-scale operations. Cluster 2 (Dark Green) represents a moderate and narrower range of employee numbers. Interestingly, Cluster 1 (Dark Blue) has the widest range of employee numbers, encompassing both relatively small and large-scale businesses. This distribution suggests that Cluster 1 may have a more heterogeneous structure composed of businesses of varying sizes. As a key indicator of business size and operational capacity, the number of employees provides an important context for understanding the segmentation between these clusters.

Another important distinction between business clusters can be observed in terms of spare part costs. Figure 9. Cluster 3 (Green) includes businesses with significantly lower spare part costs compared to the other two clusters. This suggests that businesses in this segment prioritize cost control or have a more efficient spare part management system. Alternatively, a different operational model (e.g., lower failure rates or different supply chain relationships) may contribute to these lower costs. Cluster 1 (Dark Blue) and Cluster 2 (Dark Green), on the other hand, have a wider cost range and, on average, exhibit higher spare part costs compared to Cluster 3. This suggests that businesses in these clusters may have more complex equipment fleets or follow different maintenance and repair strategies.

Box plots showing the distribution of clusters on key operational variables Figure 8 and Figure 9 indicate significant strategic and positioning differences among the companies studied.

Cluster 1 (Dark Blue): Resource-Intensive and Comprehensive Service Providers:This cluster includes companies with a wide range of employee numbers, high spare part costs, and the largest number of service vehicles. These characteristics may align with a strategy of offering more complex, customized, or geographically dispersed services. High spare part costs may indicate the use of diverse or specialized equipment, while a large number of service vehicles may signify a fast and widespread service network. Businesses in this segment may be competing in markets where quality, expertise, or rapid response are key priorities.

Cluster 2 (Dark Green): Lean and Niche-Focused Businesses: This cluster is characterized by lower number of employees and order volume. Moderate spare part costs and service vehicle counts may reflect a more balanced and likely locally focused operational structure. These businesses may focus on a specific customer segment or market niche by selectively utilizing their resources. The value they offer may include personalized service, local expertise, or customized solutions.

Cluster 3 (Green): High-Volume and Cost-Focused Businesses: Businesses in this cluster are characterized by a high number of employees and high order volume. This suggests an operational strategy focused on leveraging economies of scale. Additionally, lower spare part costs compared to other clusters may indicate that this segment has adopted cost leadership or efficient resource management principles. A smaller number of service vehicles may suggest that repair processes are organized differently (e.g., centralized workshop focus). Businesses in this cluster may serve a broad customer base in a market with high price sensitivity or demand for standardized services.

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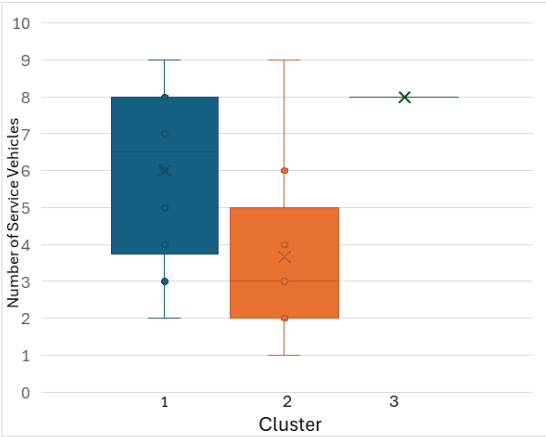

Figure 10. Clustering analysis based on the number of service vehicles

Figure 10 presents the distribution of the Number of Service Vehicles across the three identified clusters, providing clear structural distinctions among the service center groups.

Cluster 1 (Dark Blue): This cluster exhibits the highest median number of service vehicles, with the median value (represented by the line inside the box) positioned around 6.5 to 7 vehicles. This indicates that service centers grouped in Cluster 1 are generally more resource-intensive in terms of fleet size. The wide box suggests a relatively high variance in the number of vehicles within this group, ranging from approximately 2 to 9, which is further confirmed by the outliers observed at the upper extreme.

Cluster 2 (Dark Green): Cluster 2 has the lowest number of service vehicles, with a median value around 3 vehicles. The distribution is much tighter and lower than Cluster 1, indicating that these service centers operate with smaller fleets and are potentially more geographically concentrated or focused on efficiency in vehicle utilization. The data points range predominantly from 2 to 4 vehicles, with an outlier at 1 and another at 6.

Cluster 3 (Green): This cluster is highly distinct, showing a consistently high number of service vehicles, with all observed data points tightly clustered around 8 vehicles. The absence of a visible box (indicating zero interquartile range) and visible whiskers suggests that the members of this cluster are highly homogeneous regarding their fleet size, all operating with a high and specific number of vehicles.

Figure 11 illustrates the distribution of the daily completed Work Orders across the three identified service center clusters. This visualization provides insights into the operational capacity and volume differences among the groups.

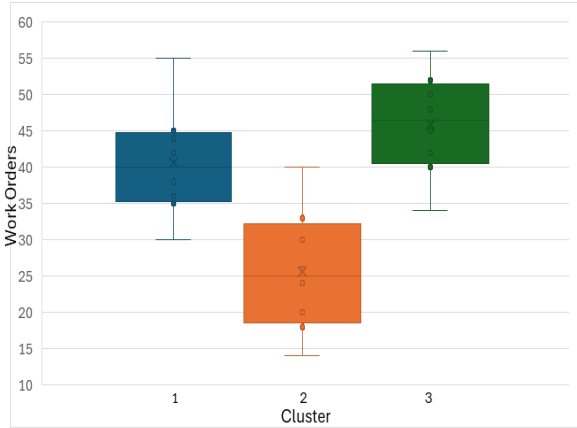

Figure 11. Cluster analysis based on work orders

Cluster 1 (Dark Blue): This cluster displays a moderate-to-high level of operational output. The median number of daily work orders (the central line in the box) is approximately 42. The interquartile range (the box itself) is relatively tight, spanning from about 35 to 44 orders. This distribution suggests that Cluster 1 service centers maintain a consistent and solid operational workflow volume.

Cluster 2 (Dark Green): Cluster 2 represents the centers with the lowest operational volume. The median work order count is significantly lower, positioned around 25 orders. The distribution shows a wide range, extending from approximately 14 to 32 orders, indicating that these centers handle fewer daily tasks compared to the other two clusters. This may reflect smaller operational scale or lower demand in their service regions.

Cluster 3 (Green): This cluster stands out for having the highest operational volume. The median work order count is around 45. Crucially, the entire interquartile range is also high, starting near 40 and extending to 50 or slightly above. This tight, high distribution indicates that centers in Cluster 3 are the most actively utilized and possess the greatest capacity for processing and completing work orders daily.

Figure 12 presents the silhouette analysis performed to evaluate the performance of the applied clustering algorithm. The graph shows the silhouette scores according to the cluster to which each observation belongs on the horizontal axis, while the clusters are grouped on the vertical axis and represented by different colors.

The silhouette score is a metric that measures the difference between an observation's similarity to its own cluster and its distance to the nearest other cluster, ranging from -1 to 1. A score close to 1 indicates that the observation fits well into its cluster; a score close to 0 indicates that the observation is similar to both clusters; and a negative score indicates that the observation is likely to be misclassified.

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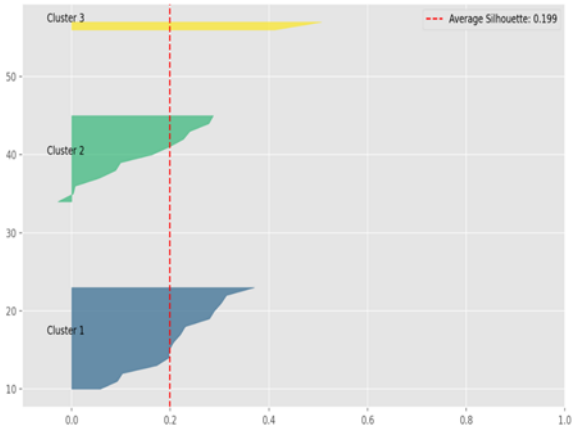

Figure 12. Silhouette Scores

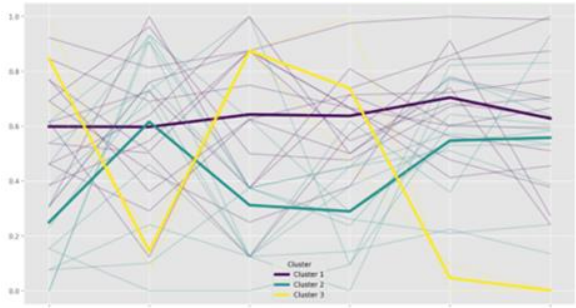

Figure 13. Parallel coordinates plot

The average silhouette score shown in the graph is 0.199. This value indicates that the distinction between clusters is limited and that the observations are not clearly separated between clusters. In other words, it can be said that the current clustering solution reflects the data structure to a limited extent.

Figure 13 presents a parallel coordinates plot illustrating the multidimensional differences among the three clusters. Each line represents a service center, while the axes correspond to the operational variables included in the analysis. The color-coded lines highlight how clusters diverge across these variables.

Overall, Cluster 1 displays a relatively stable pattern across the axes, whereas Cluster 2 tends to show lower values on several variables. Cluster 3 exhibits more pronounced fluctuations, indicating greater heterogeneity within this group. Although the clusters overlap on some dimensions, distinct separations appear on others, reflecting structural and operational differences among the service centers.

This visualization provides a comprehensive view of how performance characteristics vary simultaneously across multiple variables, offering valuable insight into the underlying patterns that define each cluster.

In the clustering analysis, the resulting segmentation provided useful insights into the structural differences among service centers. However, the overall Silhouette score was relatively low, which indicates that the clusters were not clearly separated. This outcome suggests that some service centers share overlapping characteristics across groups, reducing the distinctiveness of the identified clusters. While the segmentation still highlights meaningful performance patterns, the limited cluster validity warrants caution in interpreting these results. Specifically, the weak separation implies that managerial decisions based exclusively on these clusters may not fully capture the heterogeneity of service centers. Employing alternative clustering methods such as density-based approaches (e.g., DBSCAN, OPTICS [28]) or probabilistic models (e.g., Gaussian Mixture Models, Latent Class Analysis [29]) in future studies could provide more robust partitions and enhance the interpretability of the findings.

When examined by cluster, Cluster 1 (blue) is the cluster with the most observations, and while most silhouette scores are positive, the values are observed to be distributed in a low range. This indicates that intra-cluster similarity may be relatively weak. For Cluster 2 (green), the scores are more scattered, and some observations have very low values. This suggests that the cluster may have a limited structure in terms of internal consistency. Cluster 3 (yellow), despite containing fewer observations, exhibits a relatively high and consistent silhouette score, indicating a better-defined structure compared to the other clusters.

The results suggest that the number of clusters used (k=3) may not be the most appropriate option for the data set.

In this regard, it is recommended to try alternative numbers of clusters, apply different clustering algorithms, or re-examine the data pre-processing steps in order to improve the clustering performance. This clustering analysis shows that the examined companies do not exhibit a homogeneous structure and are divided into meaningful segments with different operational characteristics.

Identifying these segments provides a valuable framework for understanding how factors affecting business performance operate differently across groups. Future research could further explore the differences in performance scores between these clusters and the interactions of the variables identified in the beta regression analysis within these distinct segments.

3. Conclusions

This study evaluated the performance of authorized service centers in the white goods sector by employing bootstrap-enhanced Data Envelopment Analysis (DEA), beta regression, and clustering techniques. The findings revealed that average Bootstrap CCR efficiency scores ranged between 0.70 and 0.90, while Bootstrap BCC scores varied from 0.75 to 0.97. These results indicate that most centers did not achieve full efficiency, with performance constraints largely driven by unbalanced resource utilization and scale inefficiencies. For instance, Bağcılar (0.96) and Şişli-Mecidiyeköy-Fulya (0.97) achieved the highest efficiency scores, whereas Gaziosmanpaşa (0.75) and Güngören (0.78) exhibited notably lower performance.

The beta regression analysis demonstrated that the number of daily completed work orders was the most significant determinant of efficiency ($\beta = 0.0360$, $p<0.001$), highlighting the strong link between operational intensity and performance outcomes. Customer satisfaction also showed a positive, though marginally

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significant, effect (β = 0.0154, p≈0.064). In contrast, variables such as workforce size, spare parts cost, and number of service vehicles did not exert a statistically meaningful influence. Notably, an increase in employee numbers failed to improve efficiency, underscoring that qualitative aspects of human resources and process management are more critical than quantitative increases alone.

The clustering analysis identified three distinct segments of service centers: (1) resource-intensive and large-scale providers (with spare parts costs averaging around ₺4000 and extensive vehicle fleets), (2) lean and niche-focused businesses (characterized by moderate costs and relatively low order volumes), and (3) labor-intensive yet cost-sensitive operators with limited output performance. These distinctions suggest that different strategic approaches are necessary across segments, ranging from process automation and workforce optimization to cost management and localized service differentiation.

This study offers a comprehensive and methodologically sound framework for decision-makers aiming to conduct performance analysis in the service sector. The detailed causal structure provided by beta regression, along with the strategic segmentation capability of cluster analysis, introduces an innovative approach to performance evaluation studies.

For future research, several extensions are recommended. Expanding the sample size and incorporating longitudinal data would allow for the dynamic assessment of efficiency trends. Comparative applications across other industries—such as healthcare, finance, and tourism—could also provide valuable cross-sectoral insights. Moreover, alternative clustering techniques could be employed to validate segmentation results. Potential options include hierarchical clustering [28], Gaussian Mixture Models (GMM) [29], Self-Organizing Maps (SOM) [30], density-based methods (DBSCAN, OPTICS) [28], fuzzy clustering (Fuzzy C-Means) [31], and Latent Class Analysis (LCA) [29]. These methods may capture heterogeneous structures more effectively and enhance interpretability in complex datasets. Finally, integrating interaction effects into beta regression models, employing structural equation modeling, or comparing results with machine learning algorithms would further strengthen both the methodological contributions and the practical relevance of performance evaluation studies.

Declaration

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article. The authors also affirm that this article is original, has been prepared in accordance with international research and publication ethics, and does not require ethical committee approval or any special permission since the study relies solely on secondary and publicly available data.

Author Contributions

The first author supervised the research process, provided guidance on the methodological framework, and contributed to the refinement of the study. The second author, who is also the corresponding author, collected and analyzed the data, applied the DEA, Bootstrap DEA, beta regression, and clustering methods, and prepared the initial draft of the manuscript. Both authors reviewed, edited, and approved the final version of the article.

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